COMSATS UNIVERSITY ISLAMABAD

Lahore Campus

Department of Computer Engineering

CPE-415_Digital Image Processing

Assignment # 02

Resource Person

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Submitted by

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Problem 3.4:

Intensity-Slicing Transformation for Bit Plane:

An 8-bit image has intensity slicing in range 0 to 255. Each intensity in binary can be written as:

$$r = b_7 2^7 + b_6 2^6 + \cdots + b_1 2^1 + b_0 2^0$$

General Transformation:

for bit plane k with $k = 0, 1, 2, \ldots, 7$,

$$T_k(r) = \begin{cases} 255, & \text{if } \lfloor r/2^k \rfloor \% 2 == 1 \\ 0, & \text{if } \lfloor r/2^k \rfloor \% 2 == 0 \end{cases}$$

8 specific transformation:

Plane (k)	Low Range (0)	High range (255)
7	[0, 127]	[128, 255]
6	[0, 63] ∪ [128, 191]	[64, 127] ∪ [192, 255]
5	[0, 31], ∪ [64, 95] ∪ [128,]...	[32, 63] ∪ [96, 127] ∪ [160, 191] ...
4	[0, 15] ∪ [32, 47] ∪ [64, 79]...	[16, 31] ∪ [48, 63] ∪ [80, 95] ...
3	[0, 7] ∪ [16, 23] ∪ [32, 39]	[8, 15] ∪ [24, 31] ∪ [40, 47] ...
2	[0, 3], ∪ [8, 11], ∪ [16, 19] ...	[4, 7] ∪ [12, 15], [20, 23] ...
1	[0, 1] ∪ [4, 5] ∪ [8, 9] ...	[2, 3] ∪ [6, 7] ∪ [10, 11] ...
0	[0] ∪ [2] ∪ [4] ...	[1] ∪ [3] ∪ [5] ...

Problem 3.5:

(a) Effect of zeroing Lower- order:

The lower-order bit planes carry the fine gray level detail. If we zero them, many nearby intensity levels collaps to the same new value.

For example:

if we zero the lowest two bits, then

0, 1, 2, 3 → 0

4, 5, 6, 7 → 4

So,

→ The number of distinct gray levels decreases,

→ histogram bins get merge together

→ contrast in fine detail usually reduced

(b) Zeroing higher order bit planes:

The higer order bit plane contains 2 controls the large scale ~~dynamic~~ brightness and dynamic range. If we zero them, the strongest effect is severe reduction of maximum possible intensity

So,

→ The histogram is commprered toward the darker end.

→ bright intensities disappears.

→ The image become significantly darker

Problem 3.6:

why discrete histogram equalization isn't flat?

Histogram equalization does not rearrange pixels, so each output gray level receives exactly same no. of pixels. Instead it only input levels through the cummulative distribution function.

In discrete form

$$S_k = T(r_k) = (L-1)\sum_{j=0}^{k} P_r(r_j)$$

Why flatness fails:

A perfect histogram requires exactly $\frac{M \times N}{L}$ pixels at each of the L output, where $M \times N$ are dimensions of image. But discreate histogram equilization doesn't force this.

The reasons are:

→ multiple input levels maps to single output level.

→ Some output levels may receive no pixel

→ rounding of cummulative transform introduces quantization effect.

Thus the result is usually more spread out then the orignal histogram, but not perfectly uniform

Problem 3.7:

why second histogram equilization pass changes nothing.

After the first equalization pass the output intensities are already aproximately uniformly distributed over $[0, L-1]$

CDF is also nearly linear. Applying Histogram equalization again, CDF of already flat histogram is still linear. So transformation becomes identity.

Problem 3.13

Histogram of pixelwise Arthematic operation.

(a) $f+g$

Histogram hights remains the same as h_f, bin locations shift right by c

$$h(f+g).k = h(f).(k-c)$$

(b) $f-g$

Histogram heights remain the same as $h(f)$, bin locations shift left by c

$$h(f-g).k = h(f).(k+c)$$

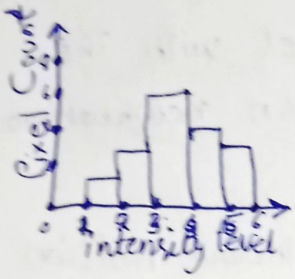
(c) $f \times g$

height remain the same as $h(f)$.
bin location are spread (multiplied) by factor c.

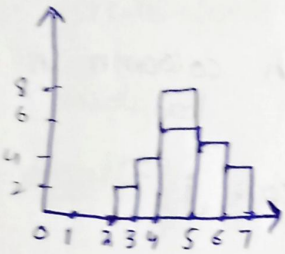
(d) $f \div g$

height remain the same as $h(f)$
bin positions are compressed (divided) by factor $\frac{1}{c}$

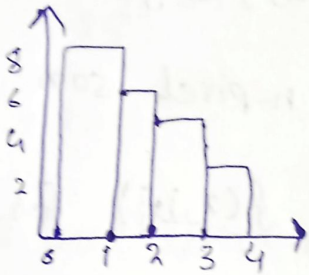
let orignal histogram:



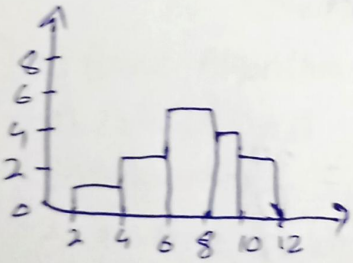
(a) f+c



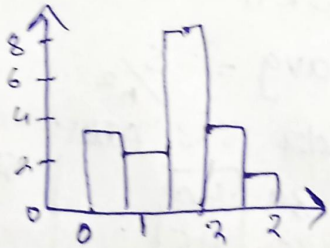
(b) f-c



(c) f×c



(d) f%c



Problem 3.15

Box filter Algorithm.

A box filter replaces each pixel with The average of all pixels in its $n \times n$ neighborhood.

(a)

1) compute full sum of first window position

$$S_0 = \sum_{i,j} f(i,j)$$

2) Store The n-pixel sum of each coloumn in The row.

$$C_j = \sum_{i=0}^{n-1} f(x,i,j) \text{ for each col. } j$$

3) for each step right:

$$S_{x+1} = S_x - C_{left} + C_{right}$$

$$avg = S/n^2$$

Example: Consider 3x3 filter:

10	20	30	40
15	25	35	45
12	22	32	42

↓ after filtering

-	-	-	-
-	22.33	32.33	-
-	-	-	-

Step 1: $C_0 = 37$

$C_1 = 67$

$C_2 = 97$

$S_1 = 201$

$avg = \frac{201}{3^2} = 22.33$

step 2:

$C_3 = 127$

$S_2 = S_1 - C_0 + C_3$

$S_2 = 291$

$avg = \frac{291}{3^2} = 32.33$

(b) computational Advantage analysis

for $n \times n$ filter on img $m \times n$:

Horizontal slide: 2 operation (subtract left + add right)

update col soms/row : n col $\times$ 2ops = 2n operation total.

Avg per pixel $\approx$ 2n operation

comparison:

	ops:	n=5	n=101
Brute-force :	n^2	25	10,201
modified :	2n	10	202

<u>result</u>:

50 x > Brute-force approch.

Problem 3.19:

median computation algorithm:

(a) Basic Algorithm:

1) Collect all n^2 pixel value from $n \times n$ neighborhood
2) Arrange values in ascending (decending) order.
3) median = value at position $\frac{n^2+1}{2}$

example:

23	45	12
67	89	34
56	78	41

1) Sorted list:

[12, 23, 34, 41, 45, 56, 67, 78, 89]

2) median = $\frac{9+1}{2} = 5 \rightarrow 45$

(b) efficient algorithm:

1) init:

Build histogram $H[0 \ldots 255]$ for first window.

2) track median:

Keep count of value $\leq$ current median

3) when sliding right:

for each value v in leaving coloumn

$H[v] \leftarrow H[v] - 1$

for each value v in entering coloumn

$H[v] \leftarrow H[v] + 1$

4) adjust median:

if removed value $\leq$ old median

decrease count

if added value $\leq$ old median

increase count

Problem 3.24:

$$\nabla^2 f = \frac{\delta^2 f}{\delta x^2} + \frac{\delta^2 f}{\delta y^2}$$ is isotropic (rotation invariant).

The laplacian operator is defined as:

$$\nabla^2 f = \frac{\delta^2 f}{\delta x^2} + \frac{\delta^2 f}{\delta y^2}$$

for the unrotated coordinates and as

$$\nabla^2 f = \frac{\delta^2 f}{\delta x'^2} + \frac{\delta^2 f}{\delta y'^2}$$

for rotated coordinate. It gives:

$x = x' \cos\theta - y' \sin\theta$

$y = x' \sin\theta - y' \cos\theta$

$\because \theta$ = angle of rotation

$$\frac{\delta f}{\delta x'} = \frac{\delta f . \delta x}{\delta x . \delta x'} + \frac{\delta f . \delta y}{\delta y . \delta x'}$$

$$= \frac{\delta f}{\delta x} \cos\theta + \frac{\delta f}{\delta y} \sin\theta$$

taking the partial derivative of expression again wrt x' yield.

$$\frac{\delta^2 f}{\delta x'^2} = \frac{\delta^2 f}{\delta x^2} \cos^2\theta + \frac{\delta}{\delta x}\left(\frac{\delta f}{\delta y}\right) \sin\theta \cos\theta + \frac{\delta}{\delta y}\left(\frac{\delta f}{\delta x}\right) \cos\theta \sin\theta + \frac{\delta^2 f}{\delta y^2} \sin^2\theta.$$

next, we compute:

$$\frac{\delta f}{\delta y'} = \frac{\delta f \ \delta x}{\delta x \ \delta y'} + \frac{\delta f \ \delta y}{\delta y \ \delta y'}$$

$$= -\frac{\delta f}{\delta x} \sin\theta + \frac{\delta f}{\delta y} \cos\theta$$

taking the partial derivative of this expression wrt y' gives.

$$\frac{\delta^2 f}{\delta y'^2} = \frac{\delta^2 f}{\delta x^2} \sin^2\theta - \frac{\delta}{\delta x}\left(\frac{\delta f}{\delta y}\right) \cos\theta \sin\theta - \frac{\delta}{\delta y}\left(\frac{\delta f}{\delta x}\right) \sin\theta \cos\theta + \frac{\delta^2 f}{\delta y^2} \cos^2\theta$$

adding two expressions for the second derivatives yields.

$$\frac{\delta^2 f}{\delta x'^2} + \frac{\delta^2 f}{\delta y'^2} = \frac{\delta^2 f}{\delta x^2} + \frac{\delta^2 f}{\delta y^2}$$

which proves that laplacian operator is independent of rotation.

Problem 3.28

Laplacian $\approx$ Unsharp Masking

Consider the following equation:

$$f(x,y) - \nabla^2 f(x,y) = f(x,y) - [f(x+1,y) + f(x-1,y) + f(x,y+1) + f(x,y-1) - 4f(x,y)]$$

$$= 6f(x,y) - [f(x+1,y) + f(x-1,y) + f(x,y+1) + f(x,y-1) + f(x,y)]$$

$$= 5\{1.2 f(x,y) - 1/5[f(x+1,y) + f(x-1,y) + f(x,y+1) + f(x,y-1) + f(x,y)]\}$$

$$= 5[1.2 f(x,y) - \bar{f}(x,y)]$$

where $\bar{f}(x,y)$ denotes the average of $f(x,y)$ in a predefined neighbourhood centered at (x,y) and including the center pixel and its four immediate neighbour.

$$f(x,y) - \nabla^2 f(x,y) \propto f(x,y) - \bar{f}(x,y)$$

The right side of equation is recognized within the Subtracting the laplacian enhances edges and fine detail

This is equivalent to unsharp masking, where

Sharpend Image = Original + Detail.